

DUBLIN AP, Ireland

WMO: 039690

Lat: 53.433N

Lon: 6.250W

Elev: 242

StdP: 14.57

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.1	29.8	24.2	18.5	30.7	26.6	20.7	32.6	36.7	49.0	34.0	48.2	7.4	260	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	71.7	62.7	69.1	61.4	67.0	60.2	64.1	69.3	62.7	67.3	61.4	65.6	11.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.9	83.6	66.6	60.6	79.7	65.2	59.3	76.1	63.9	29.4	69.4	28.4	67.4	27.5	65.7	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.9	26.6	23.8	22.2	76.0	3.2	2.6	19.9	77.9	18.0	79.4	16.3	80.9	14.0	82.8
			21.9	66.9	2.9	1.7	19.8	68.1	18.1	69.1	16.4	70.0	14.3	71.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	41.5	41.7	43.5	46.6	51.1	55.9	59.3	59.0	55.5	50.8	45.3	42.2
	DBStd	8.01	5.18	4.98	4.92	4.60	4.36	4.03	3.69	3.60	4.02	4.77	5.27	5.86
	HDD50	1331	265	237	209	122	39	3	0	0	4	48	157	247
	HDD65	5696	728	654	667	553	430	272	179	190	285	439	592	707
	CDD50	1115	2	3	6	18	74	181	289	278	170	74	15	5
	CDD65	8	0	0	0	0	0	1	3	3	1	0	0	0
	CDH74	21	0	0	0	0	0	5	10	5	1	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.9	14.0	13.5	12.9	11.6	11.3	10.5	10.0	10.4	10.9	12.0	12.6	13.2
Precipitation	PrecAvg	29.4	2.6	2.0	2.1	2.1	2.2	2.3	2.2	2.7	2.5	3.0	2.9	2.9
	PrecMax	40.6	4.7	5.0	4.2	5.8	5.9	5.8	6.5	6.6	5.7	7.1	7.4	8.5
	PrecMin	21.9	0.6	0.2	0.2	0.3	0.2	0.2	0.4	0.3	0.1	0.4	0.8	0.6
	PrecStd	3.9	1.1	1.1	1.0	1.1	1.2	1.3	1.2	1.4	1.3	1.6	1.6	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	56.2	58.7	63.3	68.7	73.8	75.4	74.2	70.9	64.2	59.4	57.1
		MCWB	52.5	52.1	51.6	55.2	59.4	61.6	64.5	65.3	62.8	59.4	56.7	54.3
	2%	DB	53.2	53.5	55.6	59.7	64.8	69.3	72.0	70.2	67.2	61.6	56.9	54.8
		MCWB	50.7	50.2	50.7	52.8	57.1	60.5	62.9	62.7	61.2	57.8	54.5	52.2
	5%	DB	51.6	51.8	53.5	56.9	61.9	66.5	69.2	67.8	64.7	60.0	55.1	52.9
		MCWB	49.3	49.1	49.6	51.1	55.2	59.0	61.4	61.2	59.1	56.5	52.7	50.5
	10%	DB	49.8	49.8	51.5	54.6	59.2	64.2	66.8	65.8	62.7	58.4	53.5	51.0
		MCWB	47.6	47.2	48.0	49.6	53.5	57.5	60.2	60.0	57.8	55.3	51.2	48.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	53.0	53.5	56.4	60.8	64.0	66.0	66.4	64.2	60.5	57.2	54.7
		MCDB	55.0	55.3	56.1	61.2	67.2	70.3	73.1	72.3	68.9	62.5	58.9	56.5
	2%	WB	51.2	51.0	51.7	53.8	58.2	61.6	64.1	64.1	62.1	58.7	55.0	52.5
		MCDB	52.9	53.0	54.5	58.4	63.4	67.3	69.9	68.5	65.9	60.9	56.5	54.2
	5%	WB	49.6	49.2	50.1	51.9	56.2	59.9	62.6	62.6	60.4	57.3	53.0	50.9
		MCDB	51.4	51.4	53.0	55.9	60.7	65.3	67.8	66.3	63.7	59.5	54.6	52.6
	10%	WB	47.8	47.5	48.4	50.1	54.4	58.3	61.1	61.1	58.7	55.8	51.5	49.0
		MCDB	49.5	49.7	51.3	53.7	58.4	63.3	65.6	64.7	61.7	58.0	53.2	50.8
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.4	12.0	13.3	13.8	14.1	13.6	13.2	12.8	11.2	9.8	9.3
		MCDBR	11.1	11.1	13.3	15.8	17.3	17.5	16.7	14.9	14.9	11.6	9.9	10.6
		MCWBR	10.3	9.5	9.6	10.2	11.1	10.6	9.9	9.0	9.9	8.9	9.0	9.9
	5% WB	MCDBR	10.9	10.8	12.0	14.8	15.7	16.2	15.2	13.6	13.4	10.6	9.6	10.6
		MCWBR	10.4	9.7	9.6	10.2	10.8	10.6	9.8	9.2	10.0	8.9	9.2	10.0
Clear-Sky Solar Irradiance	taub	0.336	0.346	0.374	0.389	0.396	0.385	0.388	0.369	0.362	0.349	0.328	0.324	
	taud	2.260	2.330	2.280	2.271	2.300	2.363	2.360	2.426	2.440	2.419	2.334	2.235	
	Ebn at Noon	194	236	254	266	271	274	271	271	258	233	198	172	
	Edh at Noon	20	26	33	38	38	36	36	32	28	23	19	17	
All-Sky Solar Radiation	RadAvg	217	440	795	1227	1559	1626	1476	1240	916	533	274	166	
	RadStd	20	37	77	116	115	161	132	107	50	44	28	19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page